

VERIFICATION TEST PLAN

ECE-593: Fundamentals of Pre-Silicon Validation
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Project Name: Design and Verification of PCIe TL using UVM

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2 Introduction:

2.1 Objective of the verification plan

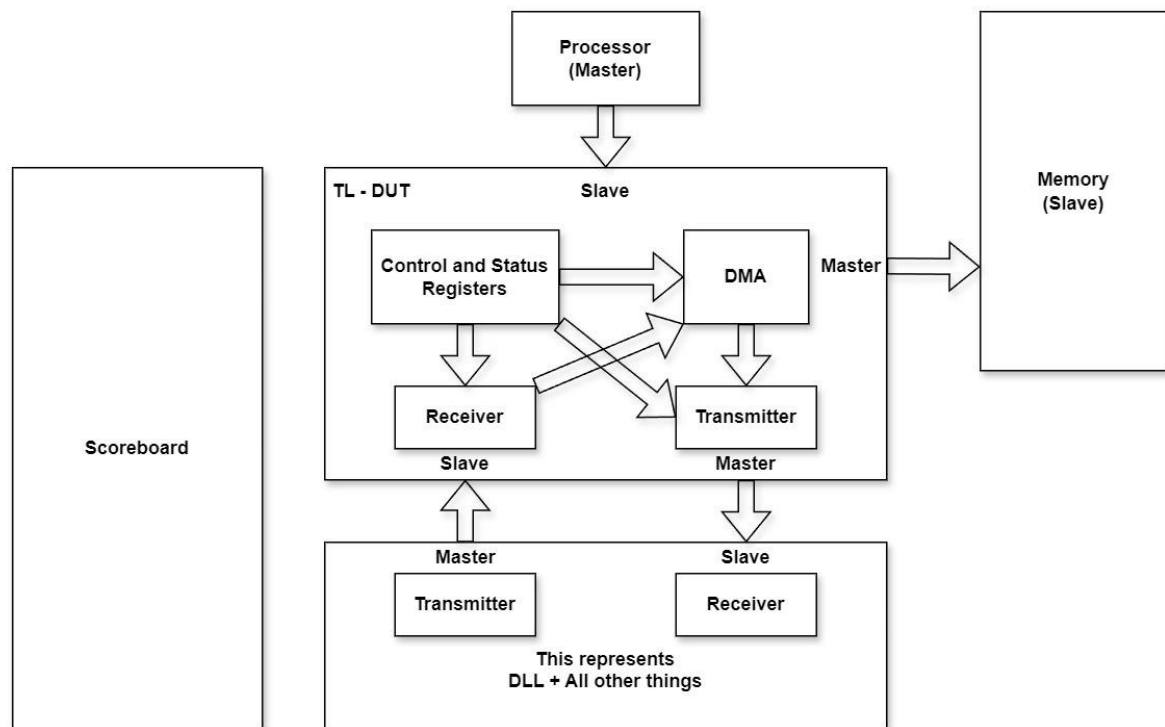
Modern electronic devices are demanding for high bandwidth. This can be achieved by (Peripheral Component Interconnect Express) PCI Express. PCI Express uses packet-based communication, packets can be generated in transaction layer.

PCI Express is used where the high speed is required for the transfer of data. Transaction Layer supports different types of transactions such as Posted and Non-Posted transactions. PCI Express uses the serial interface through which clock skews can be reduced and higher throughput can be achieved which is useful in chip verification because chip verification depends on throughput rate.

Universal Verification Methodology (UVM) is used for the verification of Transaction Layer Packet. In this, verification environment is created using four Universal Verification Component such as Advanced Peripheral Bus (APB) UVC, Advanced eXtensible Interface (AXI) UVC, Data Link Layer Transmit (DLL_TX) UVC and Data Link Layer Receive (DLL_RX) UVC. These UVC components are going to help to drive the transactions.

Design Under Test (DUT) consists of Transaction Layer Packet. Constrained random verification used for complex system. In this paper the verification is done for the PCI Express Transaction Layer to verify the transactions. Simulation results are shown in Questasim tool

2.2 Top Level block diagram



2.3 Specifications for the design

1. PCIe Transaction Layer RTL Design Specification

- Top Level Architecture
- Micro Architecture
- Interfaces
- Registers
- FSM
- Sub Blocks RTL Coding
- RTL Integration

2. PCIe Transaction Layer Test Bench Development using SV and UVM

- TB Architecture
- Test Plan
- Develop TB Components
- TB Integration
- Run Testcases, Debug
- Setup the Regression

3 Verification Requirements

3.1 Verification Levels

High Level Scope of Verification

Verify the PCIe Transaction Layer features for different use cases of PCIe End Point Devices in simulation model.

Verification includes testing of following high level items,

- Reset
- Register Programming
- Link Training
- VC0 Flow Control Initialization
- VC1 – VC7 Flow Control Initialization
- Enumeration
- Memory Write
- Memory Read
- Configuration Write
- Configuration Read
- IO Write
- IO Read
- Message Passing
- Error

3.1.1 What hierarchy level are you verifying and why?

The architecture of transaction layer, at transmit side processor will load descriptors (both transmit and receive) to external memory program, Processor is the only component which will program the descriptors, always knows where to write the data (to be used for transmit purpose) and from where to read the data (to be used for receive purpose), TX descriptor processing engine will start fetching descriptors, it is passed to DMA block then it will create a TLP. To create TLP it will get information from the configuration register to fill header fields. Then whole TLP is framed and ECRC is generated and is forwarded to transmit block. Transmit block will transmit 32 bit per clock cycle. At receive side receive block will receive the TLP frame and then it is decoded, passed to DMA receive block than it can be sent to the RX descriptor processing block. Descriptor format is 64 bit first 32 bit is used to find the address where to send TLP and second 32 bit is used find direction as either TX or RX

PCI Express Transaction Layer supports the split transaction. These are verified by using AXI UVC where write and read transactions are verified and also the targeted test cases are generated for targeted TLPs which have been verified. In this way constrained random verification of PCI Express Transaction Layer is done.

3.1.2 How is the controllability and observability at the level you are verifying?

3.1.3 Are the interfaces and specifications clearly defined at the level you are verifying. List them.

Interfaces

- Interface with the processor
 - AXI
 - TL acts as a slave for this interface
- Interface with the memory
 - AXI
- Interface with the DLL and other components till Endpoint and EP memory
 - Custom protocol

4 Required Tools

4.1 List of required software and hardware toolsets needed.

Type	Name	Manufacturer	Comment
Software	Questasim	Mentor Graphics	
Software	GVIM	Open-Source	
Software	Linux	Open-Source	
Software	UVM-1.2	Accellera	

4.2 Directory structure of your runs, what computer resources you will be using.

```
.
├── axi_intf.sv
├── pcie_tl.sv
├── run.do
├── test_lib.sv
├── tl_dll_intf.sv
├── top_tb.sv
├── transcript
└── wave.do
```

5 Risks and Dependencies

- 5.1 List all the critical threats or any known risks. List contingency and mitigation plans.

6 Functions to be Verified.

- 6.1 Functions from specification and implementation

- 6.1.1 List of functions that will be verified. Description of each function

Test Scenario#1 [Linkup indication]

Objective	To verify the functionality and robustness of the PCIe Linkup indication mechanism, ensuring that the PCIe link is correctly established, trained, and monitored as per the PCI Express specification
Pre-Condition	
Test Description	
Passing criteria	

Test Scenario#2 [TL and DLL sequence]

Objective	To verify the correct sequence and interaction between the PCIe Transaction Layer (TL) and Data Link Layer (DLL), ensuring data packet integrity, acknowledgment mechanisms, and error-free communication as per the PCIe protocol specifications.
Pre-Condition	
Test Description	
Passing criteria	

Test Scenario#3 [VC initialization]

Objective	To verify the correct initialization and configuration of Virtual Channels (VCs) in PCIe, ensuring proper resource allocation, traffic segregation, and compliance with the PCIe specification
Pre-Condition	
Test Description	
Passing criteria	

Test Scenario#4 [Generate CFG WR TLP]

Objective	To verify the correct generation of Configuration, Write (CFG WR) TLPs in PCIe, ensuring accurate packet formation, address targeting, payload handling, and compliance with the PCIe specification
Pre-Condition	
Test Description	
Passing criteria	

Test Scenario#5 [Generate CFG RD TLP]

Objective	To verify the correct generation of Configuration Read (CFG RD) TLPs in PCIe, ensuring accurate packet formation, address targeting, and adherence to PCIe protocol specifications for configuration read operations.
Pre-Condition	
Test Description	
Passing criteria	

Test Scenario#6 [Frame CFG TLP]

Objective	To verify the correct framing of Configuration (CFG) TLPs in PCIe, ensuring accurate encapsulation, alignment, and compliance with the PCIe specification for reliable communication between endpoints and configuration spaces.
Pre-Condition	
Test Description	
Passing criteria	

Test Scenario#7 [Drive the TLP]

Objective	To verify the proper transmission and handling of Transaction Layer Packets (TLPs) in PCIe, ensuring accurate data delivery, flow control, and compliance with PCIe protocol specifications.
Pre-Condition	
Test Description	
Passing criteria	

Test Scenario#8 [Enumeration]

Objective	To verify the correct execution of the PCIe enumeration process, ensuring accurate device discovery, resource allocation, and configuration as per PCIe specifications.
Pre-Condition	
Test Description	
Passing criteria	

- 6.1.2 List of functions that will not be verified. Description of each function and why it will not be verified.
- 6.1.3 List of critical functions and non-critical functions for tapeout

7 Tests and Methods

- 7.1.1 Testing methods to be used: Black/White/Gray Box.
- 7.1.2 State the PROs and CONs for each and why you selected the method for this DUV.
- 7.1.3 Testbench Architecture; Component used (list and describe Drivers, Monitors, scoreboards, checkers etc.)
Include general testbench architecture diagram and how it relates to your design
- 7.1.4 Verification Strategy: (Dynamic Simulation, Formal Simulation, Emulation etc.) Describe why you chose the strategy.

Dynamic Simulation

Dynamic simulation for PCIe Transaction Layer (TL) Verification ensures correct protocol handling, data integrity, and performance compliance under various conditions. This is done using UVM-based testbenches and dynamic stimuli generation

- 7.1.5 What is your driving methodology?

UVM:

UVM (Universal Verification Methodology) is a standardized SystemVerilog-based verification methodology used to create scalable, reusable, and efficient testbenches for functional verification of digital designs. It is widely used for verifying complex SoCs, ASICs, and FPGA designs.

- Standardized Framework – Enables industry-wide reuse and consistency
- Modular & Scalable – Encourages reusable test components.
- Object-Oriented – Uses SystemVerilog classes for flexibility.
- Constrained Random Testing – Generates randomized test cases for better coverage.
- Functional Coverage – Ensures all test scenarios are met.

- 7.1.5.1 List the test generation methods (Directed test, constrained random)
- 7.1.6 What will be your checking methodology?
- 7.1.6.1 From specification, from implementation, from context, from architecture etc

7.1.7 Testcase Scenarios (Matrix)

7.1.7.1 Basic Tests

Test Name / Number	Test Description/ Features
1.1.1	Check basic read operation
1.1.2	

7.1.7.2 Complex Tests

Test Name / Number	Test Description/ Features
1.2.1	Concurrent events (R+W) Conditions: fifo_full/fifo_empty/always_full/always empty etc.
1.2.2	

7.1.7.3 Regression Tests (Must pass every time)

Test Name / Number	Test Description/Features
1.3.1	Tests that should always pass
1.3.2	

7.1.7.4 Any special or corner cases testcases

Test Name / Number	Test Description
1.4.1	Special Case testing tests and conditions
1.4.2	Bug injection and testing scenario

8 Coverage Requirements

8.1.1.1 Describe Code and Functional Coverage goals for the DUV

8.1.1.2 Formulate conditions of how you will achieve the goals. Explain the Covergroups and Coverpoints and your selection of bins.

8.1.2 Assertions

8.1.2.1 Describe the assertions that you are planning to use and how it will help you improve the overall coverage and functional aspects of the design.

9 Resources requirements

9.1 Team members and who is doing what and expertise.

Team member	Tasks	Comments
Vamsidhar Reddy Eraganeni	Top Test Bench (top_tb.sv)	
Saketh Kokkanti	AXI Interface (axi_intf.sv) Test lib (test_lib.sv) TL DLL Interface (tl_dll_intf.sv)	
Durgaprasad Allada	PCIe TL Design (pcie_tl.sv)	

10 Schedule

10.1 Create a table with a plan of completion. You can use milestones as a guide to fill this.

Milestone#	ETA	Comments
M1	WW05	
M2	WW07.2	
M3	WW08.4	
M4	WW10.2	
M5	WW12.2	

11 References Uses / Citations/Acknowledgements

GitHub: https://github.com/vamsireddysup/team_2_PCIe_TLP

UVM User Guide:

https://www.accellera.org/images/downloads/standards/uvm/uvm_users_guide_1.2.pdf

UVM Class Reference Manual:

https://www.accellera.org/images/downloads/standards/uvm/UVM_Class_Reference_Manual_1.2.pdf

UVM Class Library Code: [UVM-1.2](#)

PCIe 5.0 Base Specification:

<https://picture.iczhiku.com/resource/eetop/SYkDTqhOLhpUTnMx.pdf>

